

Hence, intensity is also 1.

17 Ans: D

Path difference = 20 cm

The easier way is to imagine the reverse where R is the source.

In this case then, we are calculate phase difference between 2 points of the same wave that is 20 cm apart.

$$\begin{aligned}\text{Hence, using } \phi &= (\Delta x / \lambda)(360^\circ) \\ &= (20/6)(360^\circ) \\ &= 1200^\circ \\ &= 1200^\circ - 360^\circ - 360^\circ - 360^\circ \\ &= 120^\circ\end{aligned}$$

18 Ans: C